ELA

English Language Assistant

From Language Learning Support to Pattern-Sequence Discovery

Rediscovering Mendeleev's method in the age of AI.

Not just new tools - a method for finding hidden structure in complex systems.

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Original Motivation

The intermediate plateau

ELA began as a practical assistant for English learners facing the intermediate-level plateau.

- Learners know basic grammar and vocabulary.
- They understand simplified English, but authentic English remains difficult.
- Real articles, videos, lectures and podcasts contain hidden structure.
- Learners need support with grammar in real context, not only translation.

Initial aim

The original goal was to help learners understand authentic English without simplifying the original material.

Why it matters

Practical origin: English learning support for people who get stuck after the beginner stage.

The Real Difficulty

Structure behind words

The problem is not only vocabulary. The problem is the hidden relationship between linguistic and grammatical constructions.

- phrase structures
- tense / aspect / mood
- syntactic roles
- idioms
- spoken patterns
- context-dependent meaning

Therefore, a useful assistant must expose the structure of authentic input, not only translate it.

Key Challenge

Human-like linguistic notes

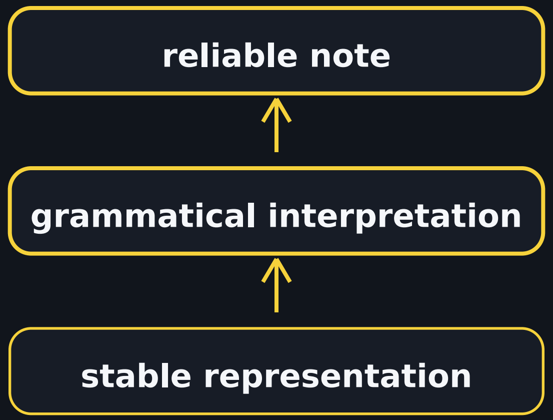
How can the system generate reliable, human-like linguistic notes for parsed English sentences?

Early problems

- Free generation sounded plausible but unstable.
- Notes often became too generic.
- Models memorised surface wording.
- Explanations were not always tied to actual grammar.
- Different sentences could share the same structure but use different words.

Conclusion

Reliable notes require a stable structural representation.



Recursive Linguistic Elements

Sentence as a tree of connected constructions

```
Sentence
 ├── Word
 ├── Phrase
 │    ├── Word
 │    └── Phrase
 │         ├── Word
 │         └── Word
 └── Phrase
      ├── Word
      └── Word
```

Core idea

Any sentence can be represented as a tree: sentences and phrases are nodes, and words are leaves.

Each element may carry

- type
- grammatical role
- part of speech
- tense / aspect / mood
- CEFR level
- source span
- linguistic notes
- translation

RLE turns a sentence into analytical units that can be analysed, validated, visualised and reused.

From RLE to Structural Patterns

Different words, same pattern

She has been working since Monday.

They have been studying for two hours.

He has been waiting all morning.

Same pattern

SUBJECT + has/have been + VERB-ing + TIME MARKER

Key idea

The transferable signal is not only in the words, but in the abstract structural pattern behind them.

In ELA, RLE nodes become analytical units and sequences of nodes form grammatical patterns.

Open and Closed Systems

Content ↔ patterns

OPEN SYSTEM

Content → Patterns → Pattern Sequences

CLOSED SYSTEM

Pattern Sequences → Inference → New Content

In an open system, content generates patterns.

In a closed system, patterns generate content.

Example: learning a new language

A learner first receives external content: words, phrases, sentences, conversations and contexts. From this content, patterns are extracted. Later, the learner uses internalised patterns to produce new sentences - this is inference from an internal pattern system.

Sequence Pattern Hypothesis

A modern AI form of Mendeleev’s method

Open phase

content → analytical units → abstract patterns → pattern sequences

Closed phase

pattern sequences → inference → new content / prediction / discovery

Mendeleev analogy

- identified analytical units
- described abstract properties
- discovered repeating patterns
- organised them into a sequence
- predicted missing elements

Core statement

Abstract pattern → sequence of patterns → prediction → discovery

ELA

Applied Meaning of the Hypothesis

From patterns to discovery

analytical units → abstract patterns → pattern sequences → inference → discovery

Mendeleev	elements → periodic patterns → missing elements
DNA	biological template → disrupted realisation patterns → candidate treatment hypotheses
ELA	Recursive Linguistic Elements → grammatical patterns → linguistic interpretation
Weather	ODE segments → regime sequences → forecasting

Evidence, Outcomes, and Demo

Two domains, one methodological logic

ELA evidence

• authentic English → RLE structure
• sentence / phrase / word constructions represented explicitly
• structural patterns support linguistic interpretation
• note generation moved toward controlled structural representation

Weather evidence

• raw weather signals → ODE-based analytical units
• transformer learned regime descriptor sequences
• 6 years → 10,055 segments
• pressure MAE 0.89 hPa at 12h
• 34% better than naive persistence

Conclusion

ELA started as a learning assistant, but the search for reliable linguistic notes led to a broader hypothesis: complex systems may be studied by discovering the abstract pattern grammar that makes real-world manifestations possible.

Next: short demo input → RLE analysis → visualisation → linguistic notes